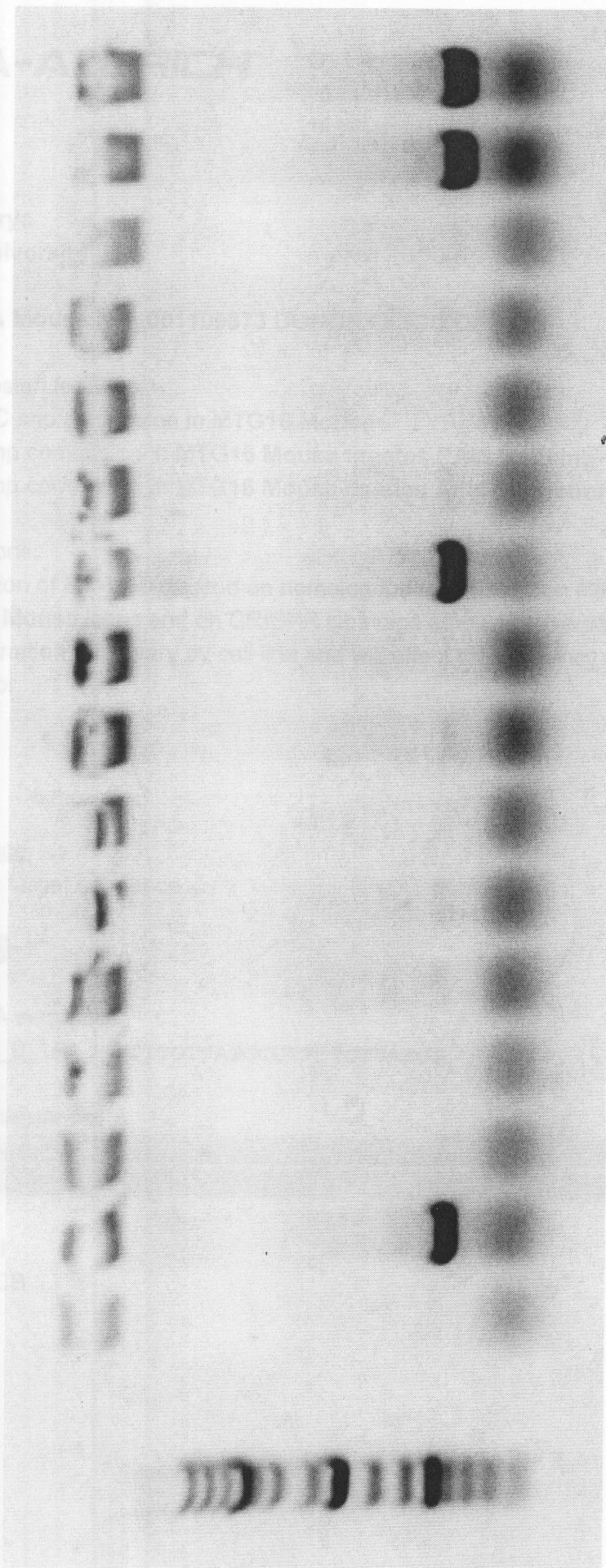


PCR screening of MTG16-F210A mutants



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MTG16 F210A Mouse NM_001109873 DONOR OLIGO DESIGN

Donor oligo design features:

- 1) **TT > GC** snp conversion in **MTG16 Mouse**
- 2) **G>A** snp conversion in **MTG16 Mouse** creates **PAM blocking**
- 3) **G>A** snp conversion in **MTG16 Mouse** deletes **AcuI** restriction enzyme

Design limitations:

- 1) Integration of SNP will depend on homologous recombination frequency at the **MTG16 Mouse** locus and on CRISPR cleavage efficiency. Homologous recombination rates vary by cell line and will affect the efficiency of integration of the donor.

KEY:

SNP

PAM blocking

Coding sequence

CRISPR gRNA target sequence

Left Homology

Right Homology

Validated gRNA sequence:

Mouse-MTG16_0_154 **TTTGTATCCCTTTTCTGAAGG**

Donor Oligo Sequence:

5' ttcatgccaaagctccaggaagccaccaactttccactgaggccggtttgttatccctGCTctg
aaAgtaagcgaaagcagcaccttcagggaacagggatggtgtacacaaaggccctgg3'

Created by: AH

Reviewed by: CB

Date: 12feb16

65

130

195

Exon5

260

Exon5

(in frame with Exon5)

P209

F210

325

390

455

480